Noise: Random Signal that impacts your message.

Modulation

↳ Amplitude modulation

→ Angle modulation
- → phase
- → frequency

Analog Communication

message signal $x(t)$

$x(t) \to$ low pass signal

Let $X(f)$ be the Fourier transform of $x(t)$

→ $x(t)$ is called low pass signal if $|X(f)| = 0 \;\forall f \notin [-W, W]$

Assumption:

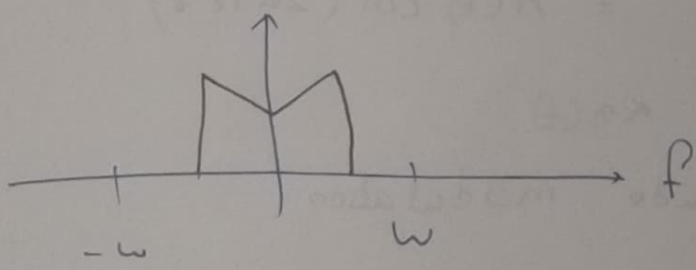
$x(t) \to$ real signal

$|x(t)| \le 1$ (W.L.O.G)

$x(t) \to$ real

↳ $X(f)$: Hermitian symmetry

$X(f) = X^*(-f)$



Bandwith of $x(t)$: W

Modulation: Process of shifting frequency

Carrier: $A_c \cos(2\pi f_c t)$

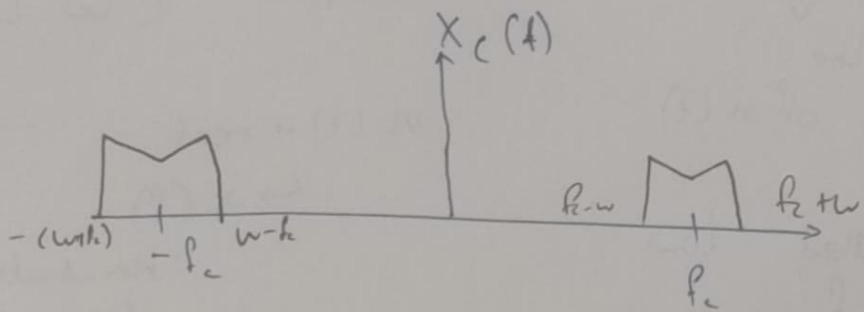
Modulatory signal:

$$x(t) \times A_c \cos(2\pi f_c t) = x_c(t)$$

$$X_c(f) = A_c X(f - f_c) \quad \text{for } f > 0$$

& $X_c(f)$ has hermitian symmetry

$= \cancel{A_c X(f}$



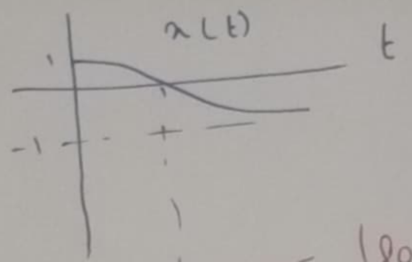
$$x_c(t) = \underbrace{(A_c x(t))}_{A(t)} \cos(2\pi f_c t)$$

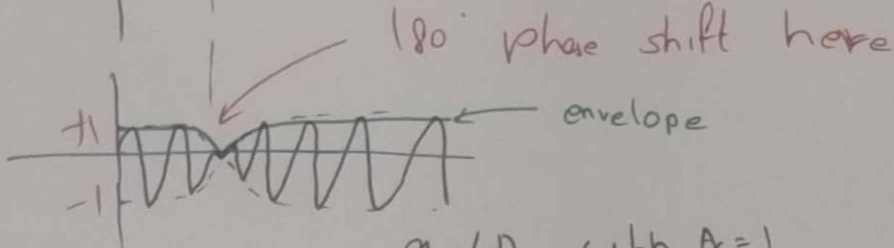
$$= A(t) \cos(2\pi f_c t)$$

$$A(t) \propto x(t)$$

↳ Amplitude modulation

$x_c(t)$ → Pass band signal with B.W $2w$, ~~carr~~ carrier f_c.



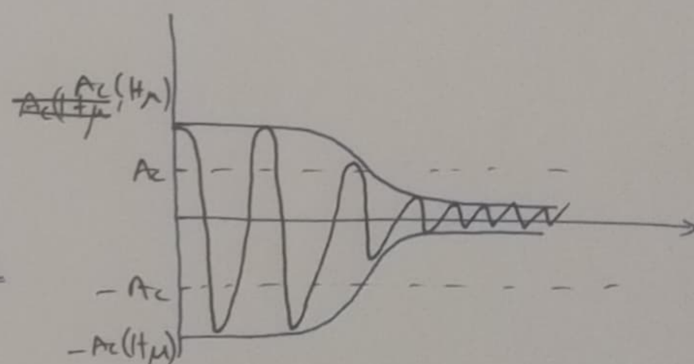


$x_c(t)$ with $A_c = 1$

$$x_c(t) = A_c(1 + \mu x(t)) \cos(2\pi f_c t)$$
$$= A_c \cos 2\pi f_c t + A_c \mu x(t) \cos(2\pi f_c t)$$

if $0 < \mu < 1$;

$$\underbrace{A_c(1 + \mu x(t))}_{A(t)} \rightarrow \text{always } \underline{\underline{+ve}}$$



$\int x(t) = 0$ {Generally}

All information is in envelope, {remove DC component}

$$X_c(f) = \frac{A}{2} X(f - f_c) + A/_2 X(f + f_c)$$

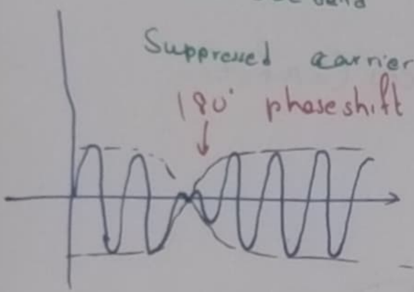
$$i.e. = \frac{A}{2} X(f - f_c) \quad ; f \geq 0,$$

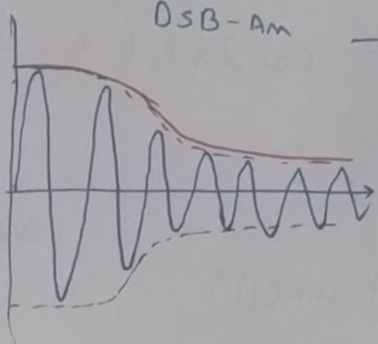
& $X_c(f)$ has hermition symmetry

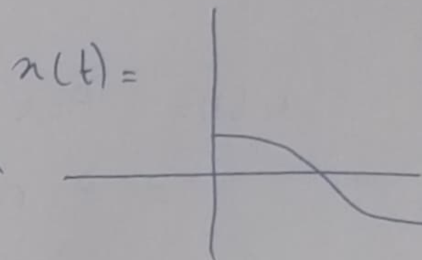
$$x_c'(t) = A(1 + \mu(x(t)) \cdot \cos(2\pi f_c t)$$

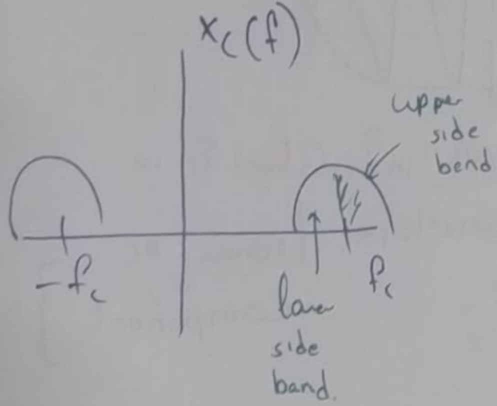
$x_c(t)$ DSB - AM - SC

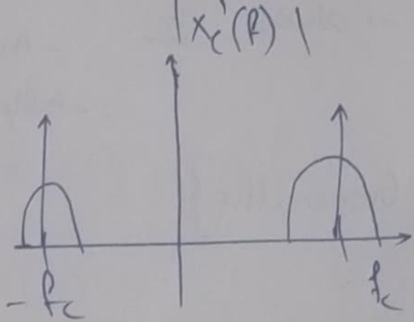
double side band











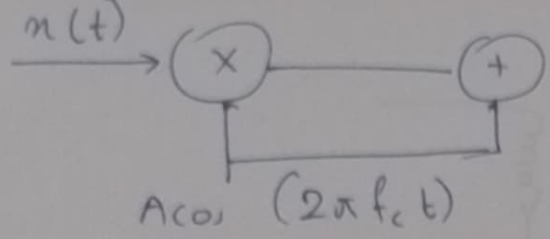
$$P_{x_c} = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} (x(t))^2 \, dt$$

$$\Rightarrow \lim_{T \to \infty} 1/_T \int_{-T/2}^{T/2} A^2 |x(t)|^2 \cos^2(2\pi f_c t) \, dt$$

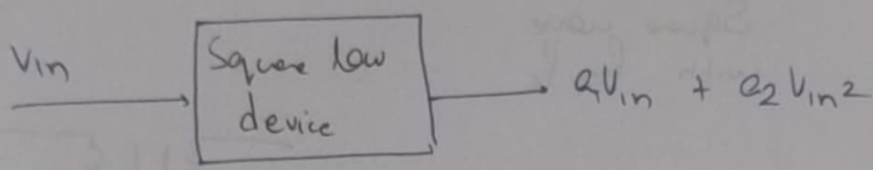
$$\Rightarrow \cancel{\frac{A^2}{2}} \frac{A^2}{2} P(x(t))$$

$$P_{x_c'}(\cancel{t}) = \frac{1}{T} \int_{-T/2}^{T/2} x(t)$$

$$= \frac{A^2}{2} + \frac{\mu^2 A^2}{2} P_x(t)$$



1) Square law device
2) Ring modulator



$a_1, a_2 \rightarrow$ constants.

Choose $\quad v_{in}(t) = x(t) + \cos 2\pi f_c t$

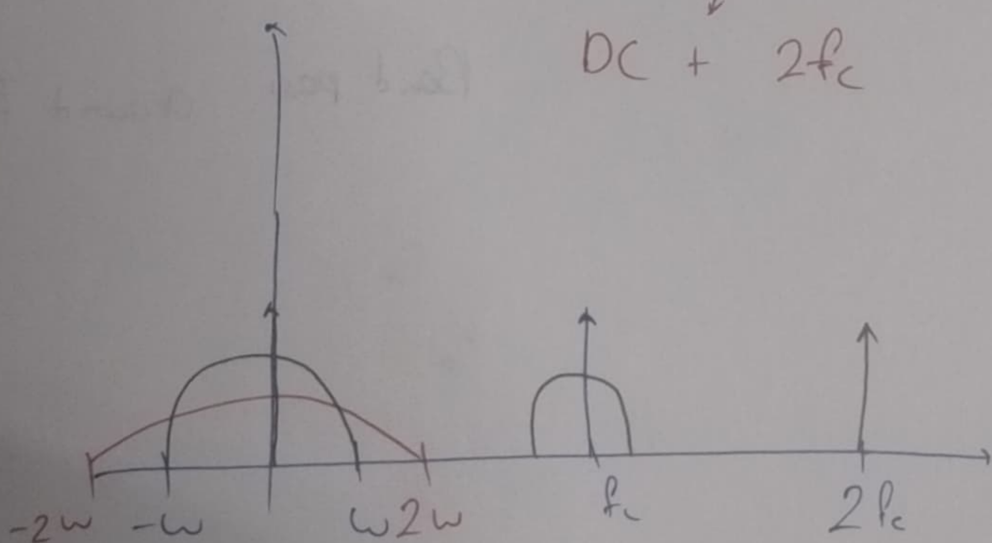
$$V_{out} = a_1 \{x(t) + \cos 2\pi f_c t\} + a_2 \{(x(t))^2 + 2x(t)\cos 2\pi f_c t + \cos^2 2\pi f_c t\}$$

Low pass W $\quad$ Band pass f_c, 2W

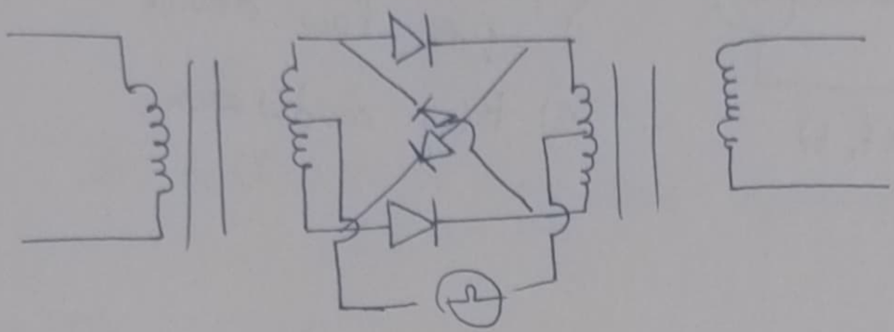
$$V_{out} = a_1 x(t) + (a_1 + 2a_2 x(t))\cos(2\pi f_c t) + a_2 \cos^2 2\pi f_c t + a_2 (x(t))^2$$

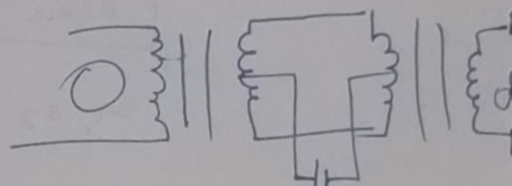
low pass 2W

DC + $2f_c$



Ring modulator:





During the half cycle: output $= x(t)$

Negative: output $= -(x(t)$

$x(t)\,c(t)$

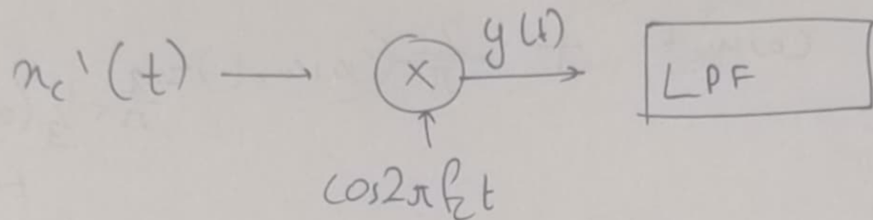
$$\frac{4}{\pi}\cos 2\pi f_c t - \frac{4}{3\pi}\cos 3\omega_c t + \frac{4}{5\pi}\cos 5\omega_c t)$$

Band pass around f_c

Demodulation.

$$x_c(t) = A(1 + \mu x(t)) \cos(2\pi f_c t)$$

$$x_c'(t) = A\, x(t) \cos(2\pi f_c t)$$



$$x'_c(t) \cos 2\pi f_c t = A\, x(t) \cos^2 2\pi f_c t$$

$$= \underbrace{\tfrac{1}{2} A\, x(t)}_{\text{Low pass signal}} \Big(1 + \underbrace{\cos(2\pi f_c t)}_{} \Big)$$

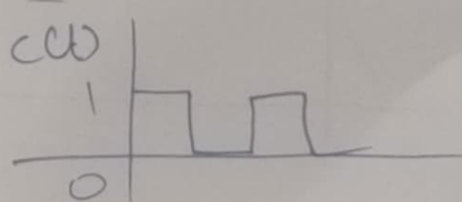
Band pass signal with carrier freq $= 2f_c$ & $BW = 2W$

$$y(t) = A\, x(t) \cos(2\pi f_c t) \cos(2\pi f_c t + \phi)$$

$$\Rightarrow \frac{A\, x(t)}{2} \left\{ \cos(4\pi f_c t + \phi) + \cos(\phi) \right\}$$

— — — —

$$x_c(t) \cdot c(t)$$

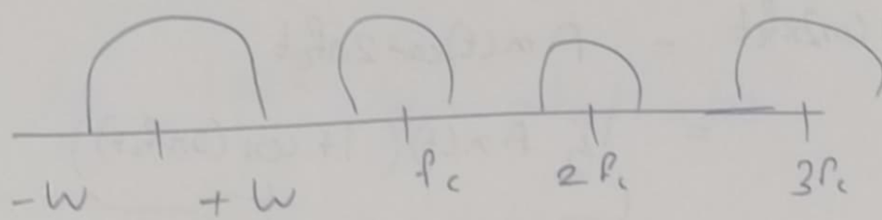


$$c(t) = \tfrac{1}{2} + \tfrac{2}{\pi} \cos(\omega_c t) - \tfrac{1}{3} \cos(3\omega_c t) + \ldots$$

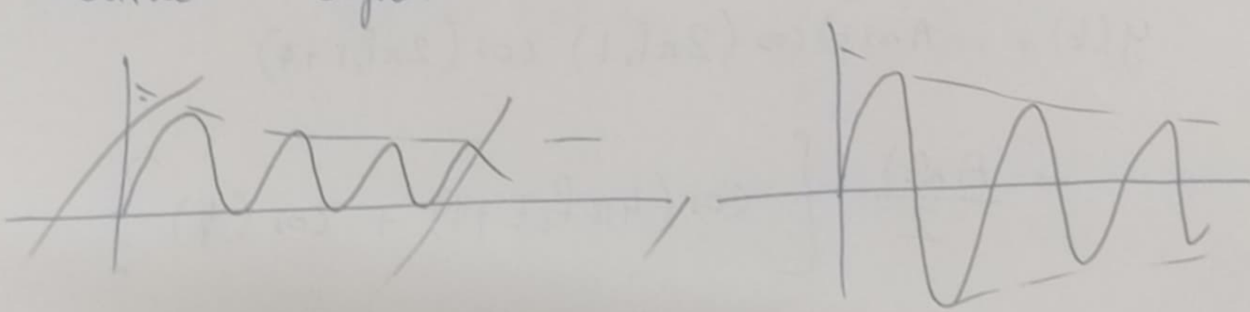
$$y(t) = x_c(t)\, c(t)$$

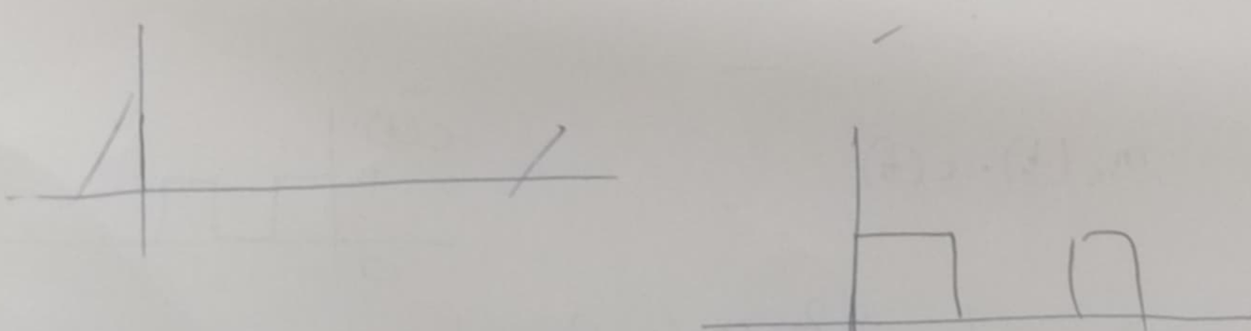
$$= \underbrace{A(1+\mu x(t))}_{A(t)} \cdot \cos\omega_c t \left[\tfrac{1}{2} + \tfrac{2}{\pi} \cos\omega_c t + \dots \right]$$

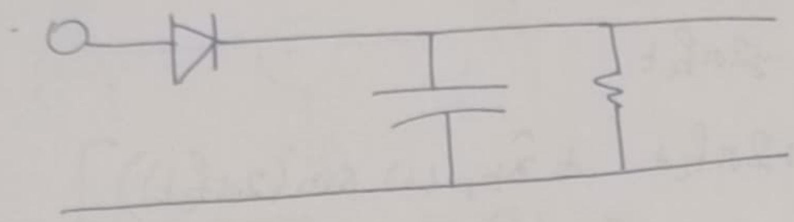
$$= \tfrac{1}{2} A(t) \cos\omega_c t + \tfrac{2}{\pi} (\cos\omega_c t)^2 - \tfrac{2}{\pi} \times \tfrac{1}{3} (\cos 3\omega_c t) + \dots (\cos\omega_c t)$$



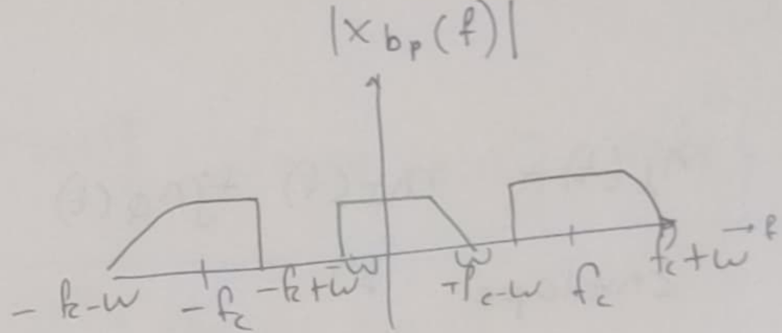
If c(t) is in syn, this will result in rectified signal.







Consider a real valued band pass signal $x_{bp}(t)$ with carrier f_c and BW $2W$



$$X_+(f) = X_{bp}(f)\, U(f)$$

unit step f^n.

$$x_+(t) = F^{-1}(X_+(f)) = F^{-1}(X_{bp}(f) \cdot u(f))$$

$$= x_{bp}(t) * \left(\frac{1}{2}\delta(t) + \frac{j}{2\pi t}\right)$$

$x_{bp}(t) * \frac{j}{2\pi t}$

hilbert transform

$$= \frac{1}{2} x_{bp}(t) + \frac{j}{\quad}$$

$$= \frac{1}{2} x_{bp}(t) + \frac{1}{2}\hat{x}_{bp}(t)$$

low pass equivalent:

$$x_l(t) = x_+(t)\, e^{-j2\pi f_c t}$$

$$x_l(t) = \frac{1}{2}\Big[\underbrace{x_{bp}(t)\cos 2\pi f_c t + \hat{x}_{bp}(t)\sin(2\pi f_c t)}_{x_I(t)}\Big]$$

$$+ \frac{j}{2}\Big[\underbrace{\hat{x}_{bp}(t)\cos 2\pi f_c t - x_{bp}(t)\sin(2\pi f_c t)}_{x_Q(t)}\Big]$$

quadrature

$$x_l(t) = x_I(t) + j x_Q(t)$$

Show that:

$x_{pb}(t) = Re\{x_l(t)\, e^{j2\pi f_c t}\}$

Envelope fn:

$$A(t) = \sqrt{x_I^2(t) + x_Q^2(t)}$$

$$\phi(t) = \tan^{-1}\left(\frac{x_Q(t)}{x_I(t)}\right)$$

If we have bandpass signal whose envelop fn is $A(t)$, then ~~ou~~ output of envelop detector is $A(t)$

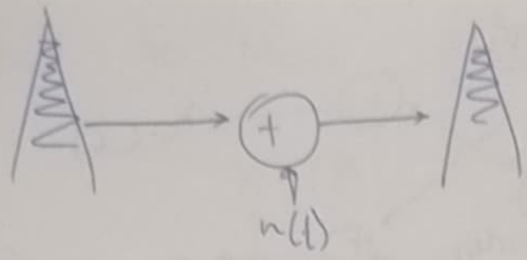
$$x_c(t) = A x(t) \cos 2\pi f_c t$$

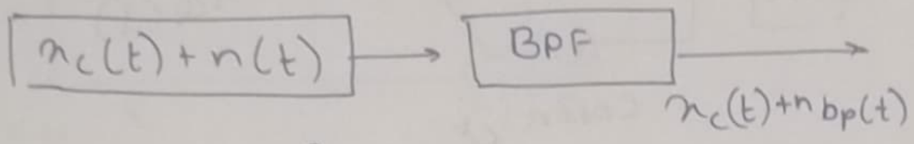
$$= Re\{A x(t)\, e^{j2\pi f_c t}\}$$

envelope( ) $= A|x(t)|$

$$x_c(t) = A(1 + \mu x(t)) \cos 2\pi f_c t$$

$$= Re\{A(1 + \mu x(t)) \cdot e^{j2\pi f_c t}\}$$



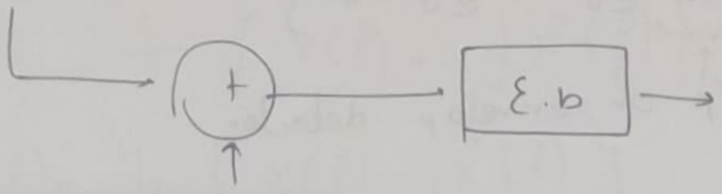


$$= Re\left\{\left[(x_I(t)+jx_Q(t))+(n_I(t)+jn_Q(t))\right]e^{j2\pi f_c t}\right\}$$

$$= Re\left\{\left[x_I(t)+n_I(t)\right]+j\left(x_Q(t)\ n_Q(t)\right)\right]e^{j2\pi f_c t}\Big\}$$

$$A(t) = \sqrt{[x_I(t)+n_I(t)]^2+[x_Q(t)+n_Q(t)]^2}$$

$$x_c(t) = Ax(t)\cos 2\pi f_c t$$

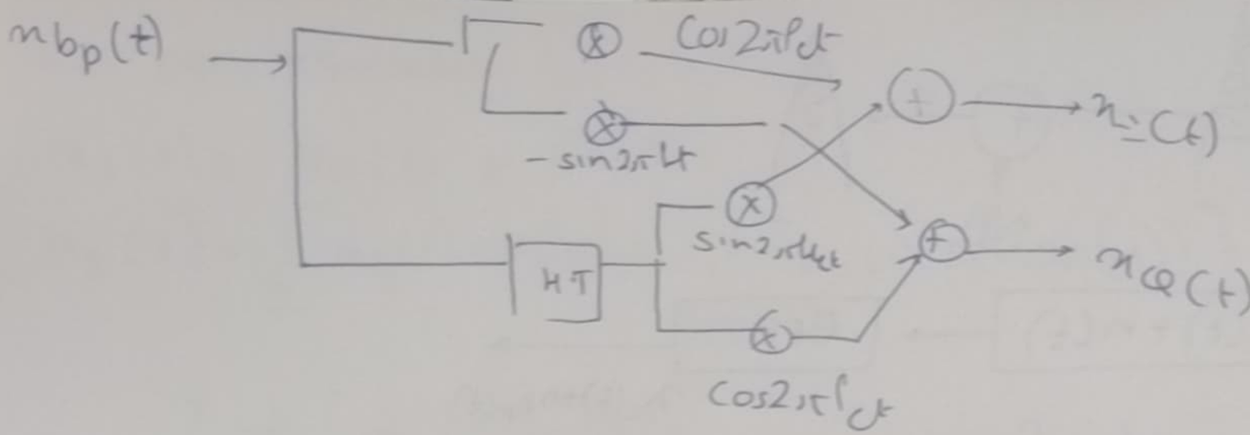


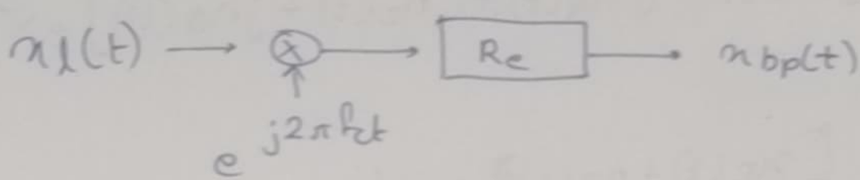
$A\cos 2\pi f_c t \quad \cos 2\pi(f_c+\Delta f)t$

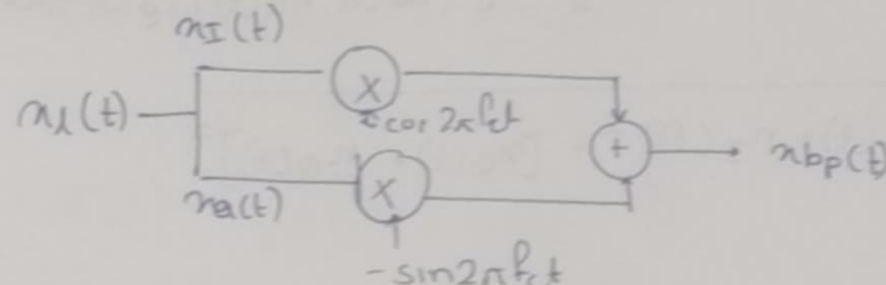
$$Re\left\{Ax(t)\,e^{j2\pi f_c t}+A\,e^{j2\pi(f_c+\Delta f)t}\right\}$$

$$= Re\Big\{\underbrace{\left(Ax(t)+A\,e^{j2\pi\Delta f}\right)}_{x_{lp}(t)}e^{j2\pi f_c t}\Big\}$$

$$x_I(t) = Ax(t)+\cancel{A\cos}\quad A\cos(2\pi\Delta f t)$$
$$x_Q(t) = A\sin(2\pi\Delta f t)$$

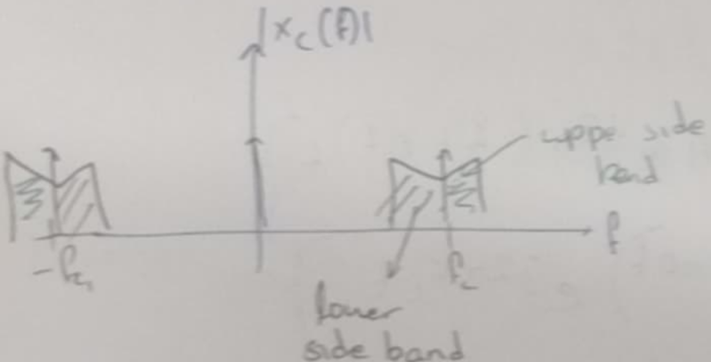




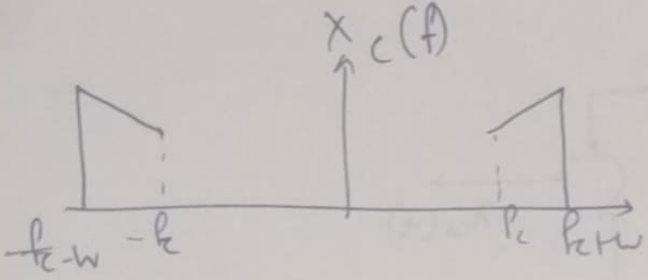


$$x_{bp}(t) = Re\Big\{\underbrace{A(t)}_{\text{o/p of envelop detector}} e^{j\phi(t)}\, e^{j2\pi f_c t}\Big\}$$

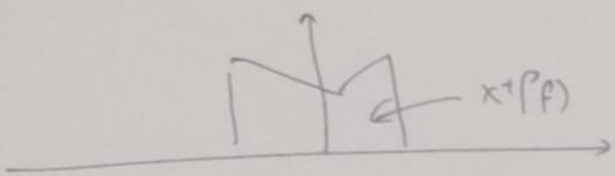
DSB AM



Signal to Noise Ratio (SNR)



Real valued signal.



$X_+(f) = X(f)U(f)$ $\qquad$ $X_-(f) = X(f)U(-f)$

$u(f) = \frac{1}{2}(1 + \text{sgn}(f))$

$U(f) = 0 \quad f<0$
$= 1/2 \quad f=0$
$= 1 \quad f>0$

$X_+(f) = \frac{X(f)}{2} + \frac{X(f)}{2}\text{sgn}(f)$

$= \frac{1}{2}\{X(f) + j\hat{X}(f)\}$

$X_-(f) = 1/2\{X(f) - j\hat{X}(f)\}$

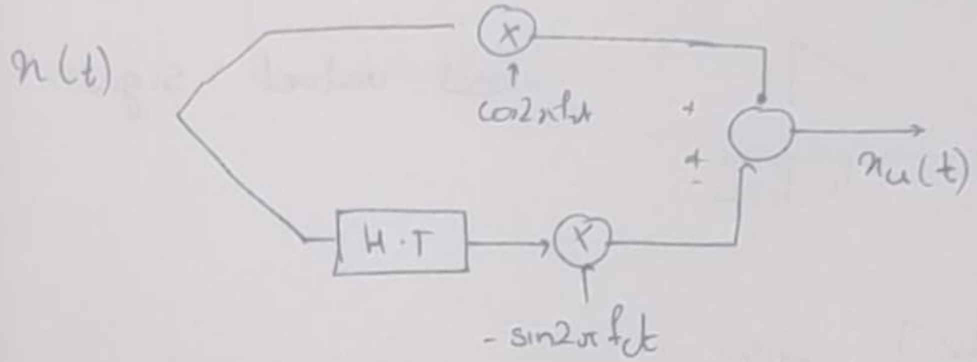
$X_u(f) = X_+(f-f_c) + X_-(\cancel{8}f+f_c)$

$= 1/2\,[X(f-f_c) + X(f+f_c)]$

$- \frac{1}{2j}[\hat{X}(f-f_c) - \hat{X}(f+f_c)]$

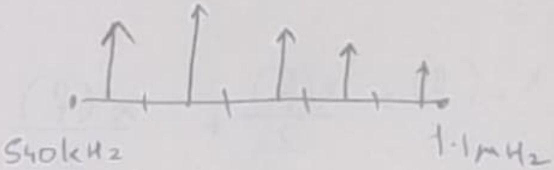
$n_u(t) = \frac{n(t)}{2}\{e^{j2\pi f_c t} + e^{-j2\pi f_c t}\} - \frac{\hat{n}(t)}{2j}\{e^{j2\pi f_c t} - e^{-j2\pi f_c t}$

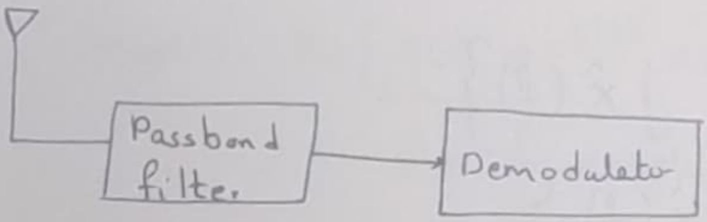
$x_u(t) = x(t)\cos 2\pi f_c t - \hat{x}(t)\sin 2\pi f_c t$



Vestigial side-band AM

AM reciever design

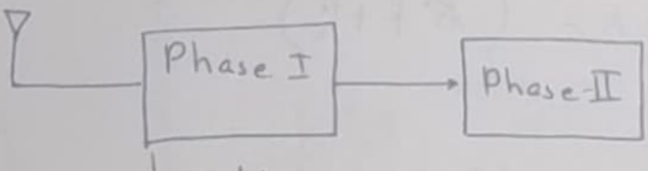




$\boxed{\text{Q-factor}}$

↳ high required.

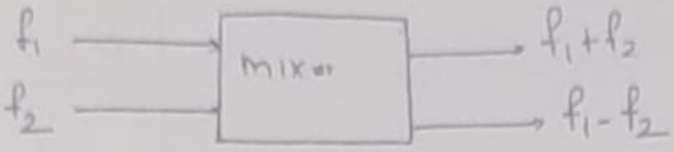
+ tunable

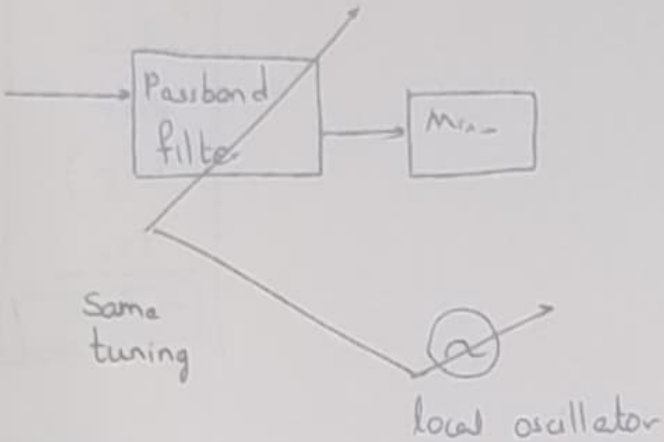


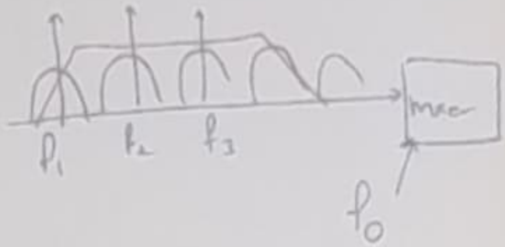
tunable

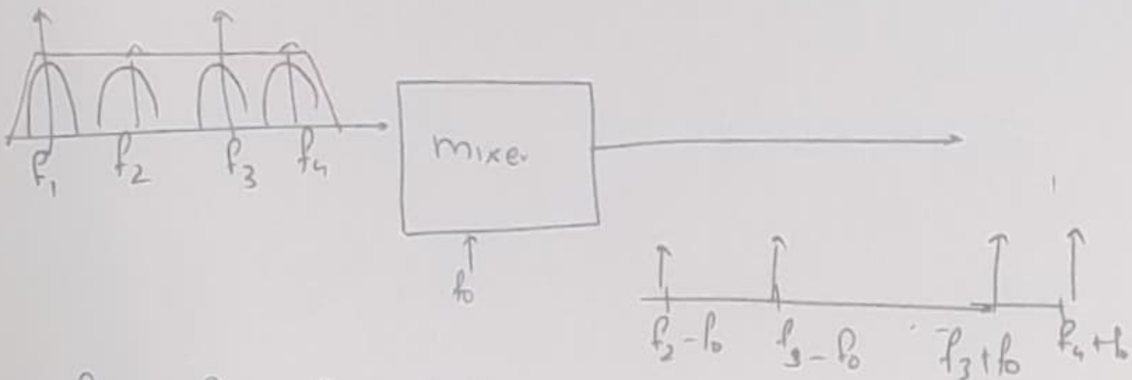
Intermediate frequency

Mixer Ckt









$= f_I$

$f_c + f_{oc} = f_I \quad \forall f_c$

$f_{oc} = f_I - f_c$

High quality BPF with carrier f_I & BW $2W$

Super-heterodyne Receiver

AM bands: 540kHz – 1.6MHz

Intermediate freq: 455kHz